

View results

Respondent

44

Anonymous

244:32

Time to complete

A. General information about the respondent

1. * Country

Czechia



2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Ministry of Education, Youth and Sports

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

<https://msmt.gov.cz/>

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

7. * Full name of officially designated contact person(s) who completed this survey

8. * Position of officially designated contact person(s) who completed this survey

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

11. Brief description of the consultation process established for the preparation of the report

We identified institutions and organisations that should develop or are developing open science and asked them to answer selected questions from the questionnaire. The list of institutions and organisations includes only those that responded.

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Czech Academy of Sciences; Charles University; Czech University of Life Sciences Prague; St. Anne's University Hospital Brno; Czech Hydrometeorological Institute; University of Technology and Economics in České Budějovice; University of Ostrava; Technology Agency of the Czech Republic; Czech Technical University in Prague; Palacký University Olomouc; Mendel University; University of Hradec Kralove; Prague University of Economics and Business; University of Chemistry and Technology Prague; Brno University of Technology

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey
(Please use commas to separate multiple links, as: link A, link B)

<https://www.avcr.cz/>; www.cuni.cz; <https://www.czu.cz/>; <https://www.fnusa.cz/>; www.chmi.cz; <https://www.vstecb.cz/>; <https://www.osu.eu/>; <https://tacr.gov.cz/>; <https://www.cvut.cz/>; <https://www.upol.cz/>; <https://mendelu.cz/>; <https://www.uhk.cz/>; <https://www.vse.cz/>; <https://www.vscht.cz/>; <https://www.vut.cz/>;

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey



Public



Private



Civil Society



Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN
SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?



Yes



No

17. 1.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In 2024, the Research, Development and Innovation Council (R&D&I Council), which is an advisory body to the government, approved new document, "Recommendations for R&D&I Financial Support Providers on the application of open science practices in the context of national programmes and public competitions in research, development and innovation". 2021 Recommendation is listed as one of the initial documents and projects on which the Recommendations for R&D&I Financial Support Providers are based.

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

- ☐ Yes
- ☒ No

19. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

- ☒ Yes
- ☐ No

20. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

Research Institutions in the Czech Republic regularly participate in Open Science Week, which is organized by the Association of Libraries of Czech Universities. <https://www.akvs.cz/en/link-to-open-access-week-2024/> Since 2023, the EOSC CZ Training Centre has provided support and facilities for training, educational and related activities. The Training Centre cooperates with research institutions in the preparation of tailor-made courses on specific topics related to Data Management and Open Science. An overview of available training in the Czech Republic is advertised on the Training centre website <https://www.eosc.cz/en/training>. There are also sources provided by Open Science centres of the universities (Masaryk University, Charles University, Silesian University in Opava, etc.) and by the National Library of Technology .

21. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

☒ Yes

☐ No

22. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

In 2024, the Research, Development and Innovation Council (R&D&I Council), which is an advisory body to the government, approved new document, "Recommendations for R&D&I Financial Support Providers on the application of open science practices in the context of national programmes and public competitions in research, development and innovation". 2021 Recommendation is listed as one of the initial documents and projects on which the Recommendations for R&D&I Financial Support Providers are based. In 2022, Managing Authority of OP JAK, an operational programme under the The Ministry of Education, Youth and Sports (MEYS), published extensive "Handbook of open science practices". Reviewed version was published in 2023 (<https://opjak.cz/dokumenty/otevrena-veda/>).

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

23. 2.1 Does your country have a national policy^[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

☒ Yes

☐ No

24. 2.1 cont. **If yes**

- please share a machine-readable PDF file of the policy using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

I tried to upload the document but this is what came up on the page: Your request is unauthorized. Your IP has been logged. Here is the link to the document:
<https://vyzkum.gov.cz/FrontClanek.aspx?idsekce=913172>

25. 2.1 cont. **If yes:**

- does it include any of the following open science elements:

- ☒ open access to scientific knowledge
- ☒ open science infrastructure
- ☐ open engagement of societal actors
- ☐ open dialogue with other knowledge systems

26. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

[i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ☐ Yes
- ☒ Partly
- ☐ No

27. 2.2 cont. **If yes or partly**, please share a machine-readable PDF file(s) of the policy(ies) using this link: <https://tinyurl.com/27p5zct8> designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

In 2024, the Research, Development and Innovation Council (R&D&I Council), which is an advisory body to the government, approved new document, "Recommendations for R&D&I Financial Support Providers on the application of open science practices in the context of national programmes and public competitions in research, development and innovation". 2021 Recommendation is listed as one of the initial documents and projects on which the Recommendations for R&D&I Financial Support Providers are based. Open Science principles are partly addressed in National Research, Development and Innovation Policy of the Czech Republic 2021+, mainly Open access to R&D results and data. OPEN DATA - Act No. 130/2002 Coll. on the support of research, experimental development and innovation from public funds and on the amendment of some related laws (applies from 1 September 2022). See § 9 Regulation of relations between the funder and the beneficiary. The most important is that the contract between the funder and the beneficiary of targeted support must include: l) the method of managing research data (so-called data management plans), m) information on the availability and dissemination of research results and research data, if they have been produced with public support under this Act, in accordance with the principle that research results and research data are not made public only in justified cases. I tried to upload the document but this is what came up on the page: Your request is unauthorized. Your IP has been logged. Here is the link to the document: (<https://www.zakonyprolidi.cz/cs/2002-130>).

28. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

[i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ☐ Yes
- ☒ Under development
- ☐ No

29. 2.3 cont. **If yes or under development**, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Until 2020, we had the Czech National Strategy for OA to Scientific Information for 2017-2020 (including publications and data) + Action Plan for its implementation. However, OA was not mandatory and this policy is not valid any more. In 2020, the Czech Government approved the National RDI Policy of the Czech Republic 2021+ as an overarching national-level strategic document for advancing all components of RDI. This document helps fulfill certain criteria of the basic conditions to be able to draw funding from EU funds in the 2021–2027 programming period (<https://vyzkum.gov.cz/FrontClanek.aspx?idsekce=913172>) There is a requirement to create an Action Plan following the Action Plan to the National Strategy for OA to Scientific Information of the Czech Republic for 2017-2020. No action plan was delivered yet.

30. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

- ☐ Yes
- ☒ Under development
- ☐ No

31. 2.4 cont. **If yes or under development**, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

The new national Open Science strategy should begin preparation in 2025.

32. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

- ☒ Yes
- ☐ Under development
- ☐ No

33. 2.5 cont. **If yes**, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

To view the elements of open science,

see: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9>

Charles University Research Data Policy (<https://openscience.cuni.cz/OSCI-42.html>) Endorsment of Open Science Principles at CZU (<https://lib.czu.cz/cs/r-16393-open-science/r-18915-open-science-na-czu>) MENDELU Open Acces and Open Science Strategy (<https://doi.mendelu.cz/pdfs/doi/9900/02/9300.pdf>) MU Open Science Strategy 2022-2028 (<https://openscience.muni.cz/open-science-na-mu/strategie-open-science-na-mu-2022-2028>) Open Science at the University of Ostrava (<https://www.osu.cz/open-science/>) Rector's Directive on Open Science – TUL (<https://www.tul.cz/veda-a-vyzkum/otevrena-veda/>) Czech Academy of Sciences - Open Science Principles of the Czech Academy of Sciences (https://www.avcr.cz/export/sites/avcr.cz/.content/galerie-souboru/Zasady_otevrene_vedy_AV_CR.pdf) TBU Guidelines - Secure Management of Research Data at Tomas Bata University in Zlín (<https://open.k.utb.cz/en/be-open/>) Strategy of the University of Economics in Prague in the field of open science (<https://knihovna.vse.cz/otevrena-veda/podpora-otevrene-vedy-souhrnne-informace/>) Open Science at the Brno University of Technology (<https://www.vut.cz/vav/vysledky/strategie-open-science-na-vut-2024-25-final-cz-pdf-p262576>)

34. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? (ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ☒ Yes
- ☐ Under development
- ☐ No

35. 2.6 cont. **If yes or under development**, please provide details and links as relevant.

Ministry of Education, Youth and Sports (MoEYS) - Johannes Amos Comenius Programme: Open Science I and II calls. Specifically, the call supports the development of a distributed National Repository Platform (hereinafter referred to as "NRP") as a key building block of the National Data Infrastructure (hereinafter referred to as "NDI"), which will create a basic environment in the Czech Republic for managing, storing, sharing and working with research data in accordance with FAIR principles, so that it is searchable, accessible, interoperable and reusable (https://opjak.cz/vyzvy/vyzva-c-02_23_014-open-science-i/). Yes, funding providers have implemented a variety of open science practices, both through their requirements and funding options (e.g. through being able to cover costs of Open Access publishing using project funds or hire data stewards as part of the project team).

36. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

☒ Yes

☐ No

37. 2.7 cont. **If yes**, please provide details and links as relevant.

All major grant providers in Czech republic have integrated OS principles into grant conditions to some extent - Czech Science Foundation (<https://www.gacr.cz>), Technology Agency of the Czech Republic (<https://www.tacr.gov.cz>), Agency for Health Research of the Czech Republic (<https://www.azvcr.cz/>), The Ministry of Education, Youth and Sports (<https://www.msmt.gov.cz>)

38. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

☐ Yes

☒ No

39. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Not now, but we expect this topic to be included in the new strategy document.

40. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ☒ Yes
- ☐ Under development
- ☐ No

41. 2.9 cont. **If yes or under development**, please provide details and links as relevant.

Open Access to the publications is monitored via Czech national CRIS (IS VaVal) - Information Register of R&D results (RIV) <https://www.isvavai.cz/>. The information about the form of access to the publication (open/embargo/restricted) is mandatory for all journal articles.

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

42. 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

In relation to gross domestic product, there will be a year-on-year decline in this expenditure, from 1.89% in 2022 to 1.83% in 2023 (Note: These figures are in line with the latest results of the CSO's extraordinary revision of the national accounts of 28 June 2024. They differ from the figures in this article, which were published before the revision). The aforementioned decline is thus already in its third year - i.e. since 2020, when the share of research expenditure in GDP reached 1.95%. (viz <https://vedavyzkum.cz/politika/politika-vyzkumu-a-vyvoje/vydaje-na-vyzkum-a-vyvoj-v-roce-2023-stouply-na-140-miliard-korun-v-pomeru-k-hdp-vsak-klesly-na-1-83>)

43. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

In the second quarter of 2022, 85% of Czech households were connected to the internet. (<https://statistikaamy.csu.gov.cz/k-internetu-jsou-pripojeni-skoro-vsichni>)

44. 3.3 Are there national research and education networks (NRENs)[i] active in the country? (ref.: (iii) 18.c)

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

☒ Yes

☐ No

45. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

CESNET is an association of universities of the Czech Republic and the Czech Academy of Sciences. It operates and develops the national e-infrastructure for science, research and education which encompasses a computer network, computational grids, data storage and collaborative environment. E-infrastructure provides services to transfer, process and store data for the needs of Research and Educational community (<https://www.cesnet.cz/en>).

46. 3.4 Are there national or institutional open science infrastructures[i] in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.



Yes



Under development



No

47. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:



federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage



community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society



platforms for exchanges and co-creation of knowledge between scientists and society



community-based monitoring and information systems to complement national, regional and global data and information systems

48. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

e-INFRA CZ stands as a unique e-infrastructure (non-commercial) dedicated to research and development in the Czech Republic. It provides a transparent environment offering comprehensive capacities and resources for the transfer, storage and processing of scientific data to all entities engaged in research and development, regardless of the scientific domain. The core activities of e-INFRA CZ as a whole gradually shift towards adhering to the FAIR principles. The following steps in the national EOSC development plan involve great number of research institutions, which will together implement so-called National Data Infrastructure with National Repository Platform in its heart. Together with this infrastructure for repositories, several EOSC-related services will be implemented, encompassing solutions for PIDs, AAI, Data Management Planning, scientific data collaboration platforms, and more. At the same time, this infrastructure will become the inherent part of the national IT research environment provided by the e-INFRA CZ. (<https://www.e-infra.cz/en>).

49. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

☒ Yes

☐ No

50. 3.5. cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

CEITEC Nano Infrastructure has reservation system with automated data storage and data.description (<https://openscience.muni.cz/en/open-science-at-mu><https://nano.ceitec.cz/>). Czech Literary Bibliography Research Infrastructure - The Data Management Plan (DMP) provides information about the lifecycle of data as it is generated, collected, processed, stored, and shared within the Czech Literary Bibliography Research Infrastructure. (<https://clb.ucl.cas.cz/en/research-data-management/>). A recognized data-management-plan(ning) supporting tool Data Stewardship Wizard (DSW) is developed in Czech Republic. This tool has increased in popularity of its use in the Czech Republic and other countries and is recommended by the Horizon Europe Programme Guide (<https://ds-wizard.org/>) please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

51. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

☐ Yes

☒ No

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

52. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

☒ Yes

☐ No

53. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

Since 2023, the EOSC CZ Training Centre has provided support and facilities for training, educational and related activities. The Training Centre cooperates with research institutions in the preparation of tailor-made courses on specific topics related to Data Management and Open Science. An overview of available training in the Czech Republic is advertised on the Training centre website <https://www.eosc.cz/en/training>. There are also sources provided by Open Science centres of the universities (Masaryk University - <https://openscience.muni.cz>, Charles University - <https://openscience.cuni.cz/OSCI-1.html>, Silesian University in Opava - <https://www.slu.cz/>, etc.) and by the National Library of Technology - <https://www.techlib.cz/cs/84873-open-science>. There is also increasing Data Steward community - the data steward, a newly established role in research data management, plays a crucial part in ensuring the quality and integrity of data throughout the research cycle. This includes the responsible handling of data, from its creation and preparation to use and storage to archiving and sharing or reuse. The data steward's role is to ensure that data management is carried out in accordance with defined principles, policies and rules that support data quality and integrity.

54. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

☒ Yes

☐ No

55. 4.2 cont. **If yes**, please provide more information and links as relevant.

Given the somewhat fragmented environment in Open Science, capacity building is currently in the hands of individual institutions. See above.

56. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)



Yes



No

57. 4.3 cont. **If yes**, please provide more information and links as relevant.

Individual institutions prepare training schemes independently. For example Palacký University Olomouc organises a series of ten modules delivered as part of the whole-life training offered to all staff and PhD students (content includes research data management, introduction to open science, commercialisation of research, ethics of scientific work, etc.), which is held twice a year. Link: <https://openscience.upol.cz/zprava/clanek/kurzy-otevrene-a-zodpovedne-vedy-jiz-potreti>. Moreover, formal and informal communities of e.g. data-stewards and open science specialists (which are active e.g. on social networks - Discord, or under the aegis of the Association of University Libraries or the Czech Association of Research Managers and Administrators) are a certain link and a possible germ of a national approach. Charles University is gradually incorporating open science competencies in educational programs. The most recent example is the MSCA COFUND project Charleston (<https://cuni.cz/UKEN-2185.html>) aimed at postdoctoral researchers includes training courses based on the ResearchComp (The European Competence Framework for Researchers). This training is expected to be extended for doctoral students as well. Charles University is also actively involved in a series of webinars "Open for you!" (<https://4euplus.eu/4EU-273.html>) which are prepared for all member universities across 4EU+ university alliance. Beyond these examples, the University Open Science Support Centre organises ad hoc trainings for researchers as well as for research support staff. University of Chemistry and Technology Prague has for Bc. students: elementary information regarding copyright, Open Access and basic OS elements (DOI, ORCID, ROR) is included as part of the „Chemical Informatics“ course (mandatory course).

58. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ☐ Yes
- ☒ Partly
- ☐ No

59. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ☒ Yes
- ☐ No

60. 4.5 cont. **If yes**, please provide more information and links as relevant.

Already in 2019, the Technology Agency of the Czech Republic funded the project TL02000013 "Quality Assurance of Open Educational Resources: how to create and use freely available learning materials properly" (2019-2021). The project developed a) a certified methodology, which is a methodological tool for creating Open Educational Resources suitable for university studies. In addition, a methodological portal was developed, which summarises the methodological knowledge for the creation of OER through a series of materials on the topic (texts, videos, images) and also provides an interactive space for users. Based on the methodological knowledge, a freely available online course (MOOC) was also created for the staff of Charles University and the general public. The methodological portal and the online course will be available after the end of the project to promote critical evaluation of sources as a defence against misinformation. A very useful national communication channel is the portal <https://vedavyzkum.cz/>, where open science appears ad hoc and rather marginally. At our university we have a position of science popularizer, who takes a conscious approach to science communication in line with open science value and principles. National Library of Technology is organizing Doctoral Summer School which focuses on topics of Open Science, Communication in Research, Career management and similar topics: <https://www.techlib.cz/en/84767-doctoral-summer-school-2023>. There are initiatives to support the dissemination of scientific results among the general public and other researchers. For example, there is a national event hosted annually called "Researchers' Night" during which participating institutions open their doors to the public and educate visitors on various aspects of the fields of research they offer. The field of open science is gaining importance in academia and university libraries are often the institutions that address and engage with open science topics in the field of education (students, academic and technical-economic staff) and science and research. The role of the Association of University Libraries of the Czech Republic is to anchor libraries in the university environment: a more active role in supporting study/research and greater involvement in key university processes. The Association is involved in Open Science initiatives. Already in 2010, the Open Access Initiative of the Czech Republic was established within the Association, as an informal group of academic librarians who have long been working on the issue of open access to scientific information and building open repositories. The importance of the OA Initiative was confirmed by the Association signing the Berlin Declaration in 2012 and by its membership in the international organisation SPARC Europe (since 2015). 2021, then the Open Science Working Group (OS WG) was established, which continues the work of the Open Access Initiative and aims to monitor all relevant trends in the field of open science, and , for example, organizes the Open Science Week (3 days of practical demonstrations, experience sharing and lively discussions).

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

61. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c, 20i)

☒ Yes

☐ No

62. 5.1 cont. **If yes**, please provide more information and links as relevant.

One of the indicators in the new methodology for the evaluation of research organisations in the higher education segment (Methodology 25+) concerning the sustainability and resilience of RPOs is the concept of Open Science (https://msmt.gov.cz/uploads/Hodnoceni_2025/Metodika_VS2025/Metodika_VS2025_.pdf). National Recovery Plan: EXCELES Programme The EXCELES programme of the National Recovery Plan emphasises open science. Procurement documents in Section 4. Subject Matter states that grantees should share research results (both publications and research data) in accordance with the well-known "as open as possible, as closed as necessary" rule, i.e., as open as possible, closed as necessary. The Technology Agency of the Czech Republic supports applied research. In the KAPPA Programme, funded by the EEA and Norway Grants, there is an obligation to ensure open access for all scientific articles resulting from a research project, in the form of Green Open Access or Gold Open Access. Each research project funded under this programme must also prepare its own data management plan. The Czech Science Foundation thus supports open access publishing on a voluntary basis. Beneficiaries are obliged to develop a research data management plan.

63. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

☐ Yes

☒ No

64. 5.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

here is no generally applicable framework that recommends this. However, open science as such, its principles and practices are developing and can be expected to find their way into career evaluation systems etc.

65. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

☒ Yes

☐ No

66. 5.3 cont. **If yes**, for each initiative, please provide more information, including on the stakeholders[i] that are involved or lead the initiative, and links as relevant.

[i] For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

One of the indicators in the new methodology for the evaluation of research organisations in the higher education segment (25+ methodology) concerning the sustainability and resilience of RPOs is the concept of Open Science. For example, at the end of 2024, the Czech Academy of Sciences approved a new strategic document, the Principles of Open Science of the CAS, which represents the basic framework and recommendations for the implementation of the principles of open science at CAS institutions. The CAS, through its institutes, has long been building and implementing open access to scientific information generally adhering to standards of scientific integrity. The CAS sees this as a path to greater quality, transparency, sustainability and reproducibility of research. The CAS also welcomes the fact that the sharing of scientific data and other digital objects in accordance with the FAIR principles (Findable-Accessible-Interoperable-Reusable) is becoming a new reality and standard in research. The CAS subscribes to the principles of Open Science in line with its commitment to disseminate research findings as widely as possible and in accordance with UNESCO recommendations and the anchoring of Open Science in national, European and international science policy. The need for greater openness of science and wider access to scientific results and data is also mentioned in the Code of Ethics for Researchers of the CAS. (<https://openscience.lib.cas.cz/open-science/institutional-policies/>) In 2022, Brno University of Technology became a signatory of the Agreement on Reforming Research Assessment and a member of the CoARA coalition. The CoARA Action Plan for BUT was also developed. <https://www.vut.cz/en/rad/evaluation>; Charles University takes an active role in initiatives aiming to reform research assessment and evaluation - it is one of COaRA members and seeks to implement its principles into internal evaluation schemes (see Charles University COaRA Action Plan 2024: <https://zenodo.org/records/12545583>). Palacký University Olomouc signed the Agreement on Reforming Research Assessment (ARRA) in July 2023, thereby committing to the need to change the culture in the R&D area and R&D evaluation, which is still overly focused on metrics, inappropriately in at least some research fields. Although some necessary steps towards this change have already been taken before joining ARRA (i.e., the first and comprehensive international peer-review evaluation based on self-evaluation in 2023, work on a strategy for Open Science and responsible Research data management is in the process thus far), we feel the philosophy of ARRA needs to be declared officially and further developed and anchored at this institution. Ministry of Education, Youth and Sports (MoEYS) - Johannes Amos Comenius Programme: Immediate and complete OA, CC licenses, DMP, FAIR data, OS procedures as part of the evaluation criteria: Call: Research Excellence Call: Call for proposals to improve results in research, development and innovation, called: Cross-sectoral cooperation Call: Social Sciences and Humanities: Man and Humanity in Global Challenges National Recovery Plan: EXCELES Programme The EXCELES programme of the National Recovery Plan emphasises open science. Procurement documents in Section 4. Subject Matter states that grantees should share research results (both publications and research data) in accordance with the well-known "as open as possible, as closed as necessary" rule, i.e., as open as possible, closed as necessary.

67. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.



Yes



No

68. 5.4 cont. **If yes**, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

It is mostly the increase in costs for publications, lots of publications in predatory or potentially predatory journals. The phenomenon of predatory journals still appears here and there, but the bigger problem from our point of view seems to be the questionable practices of some established publishers (increase of special issues in journals, high APCs, short and questionable peer review process). There are initiatives by some institutions and universities to discuss this phenomenon and to warn authors about possible pitfalls. A useful initiative is <https://www.stoppredatorypractice.com>. On the level of individual researchers, Open Science is often felt to be a lot of work for little gain, especially as the overall funding of research is considered to be rather low so the additional costs of implementing open science practices (APCs, data stewards...) are often viewed as a waste of resources. Moreover, the evaluation schemes don't take these additional efforts into consideration yet. For example improving research data management requires not insignificant amount of resources (time, trainings, infrastructure) which are difficult to get by and researchers are rarely evaluated for the additional efforts. Regarding scholarly communication, Czech researchers are conservative in viewing highly impacted journals as the pinnacle of publishing and it is difficult for them to accept anything different than traditional, highly impacted journals as the right venue to publish best research. Moreover, some Czech funders are still rewarding high impact publications which, in combination with open access mandate (e.g., in Czech P JAC funding scheme), results in a significant increase of publication costs (e.g., APC costs for Nature publications). Additionally, data migration can present challenges, especially when moving research data between platforms or repositories. This process can sometimes result in data loss or mismanagement if improperly handled. The lack of interoperability between different data platforms can also hinder the efficient sharing and reuse of research data, which is essential to open science principles. Another concern is the privatisation of research data, as some private companies may attempt to control access to data initially created as part of publicly funded research. This could result in the exploitation of publicly funded knowledge, where data or research outputs are not made available for broader societal benefit but are locked behind paywalls or commercial interests. High article processing charges (APCs) associated with some open access models remain a significant barrier for many researchers, especially in disciplines with less funding. These costs can disproportionately affect scientists from less affluent institutions or countries, exacerbating global inequalities in access to publishing opportunities and research dissemination.

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

69. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

- ☐ YES, at the national level
- ☐ YES, at the institutional level
- ☐ YES, at both the national and institutional levels
- ☒ No

70. 6.1 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

For the time being, we have no information about similar initiatives or activities.

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

71. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

In 2010, the Open Access Initiative of the Czech Republic was established within the Association of Libraries of Czech Universities, which has long been working on the Open Access to scientific information and Open Repositories. The importance of the OA Initiative was confirmed by the Association signing the Berlin Declaration in 2012 and by its membership in the international organisation SPARC Europe (since 2015). In 2021, then the Open Science Working Group (OS WG) was established, which continues the work of the Open Access Initiative.

72. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

☐ Yes

☒ No

73. 7.2 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The new national Open Science strategy should begin preparation in 2025.

74. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

☐ Yes

☒ No

75. 7.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The Czech Republic is promoting Open Science principles through initiatives and funding mechanisms described in previous points. However, this environment is fragmented and it is necessary to unify this topic at national level.

Part 8. GENERAL CONSIDERATIONS

76. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

☐ Yes

☒ No

77. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org